VHDL NOT Gate Simulation

Questa + Logisim-evolution Workflow

Assignment	VHDL NOT Gate
Software	Questa FPGA Starter Edition / Logisim-evolution
Date	2026-10-06

1. Objective

The purpose of this assignment is to become familiar with the basic VHDL workflow and the use of Questa and Logisim-evolution. A one-input NOT gate was implemented in VHDL, imported into Logisim-evolution, and simulated to verify the expected truth-table behavior.

2. VHDL Implementation

The design has one input **a** and one output **y**. The statement `y <= NOT a;` implements the logical inversion.

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY test IS
    PORT (
        a : IN std_logic;
        y : OUT std_logic
    );
END test;

ARCHITECTURE TypeArchitecture OF test IS
BEGIN

    y <= NOT a;

END TypeArchitecture;
```

3. Logisim-evolution Circuit and Simulation

The VHDL entity was placed in the Logisim-evolution circuit and connected between an input pin **a** and an output pin **y**. The circuit was then tested for both possible input values.

Figure 1. Simulation result for a = 0

When a = 0, the simulated output is y = 1.

Figure 2. Simulation result for a = 1

When a = 1, the simulated output is y = 0.

4. Results

Input a	Output y	Expected	Result
0	1	1	PASS
1	0	0	PASS

The measured outputs match the expected NOT-gate truth table: the output is the logical complement of the input in both test cases.

5. Conclusion

The NOT gate was successfully implemented in VHDL and simulated in Logisim-evolution using the Questa backend. The simulation verified that **a = 0 produces y = 1** and **a = 1 produces y = 0**. Therefore, the implementation behaves correctly as a NOT gate.

Submission Files

File	Purpose
not_gate.vhd	VHDL source code
not_gate.circ	Logisim-evolution project
NOT_Gate_Report.pdf	PDF report